

Lidar Semantic Global Localization (LSGLOC) Demo

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Problem Statement

Robots currently **cannot autonomously determine their initial pose or continuously localise themselves** within the 2D map coordinate frame without manual assistance or prior environment mapping.

Problems

- **Global Localization:** No manual initial pose is given to the robot during deployment.
- **No Prior Mapping:** The robot must make use of a CAD floor plan that describes the geometry of the environment.
- **Long-Term Localization:** A robot must continuously localize within an environment to allow waypoint navigation.

Solution

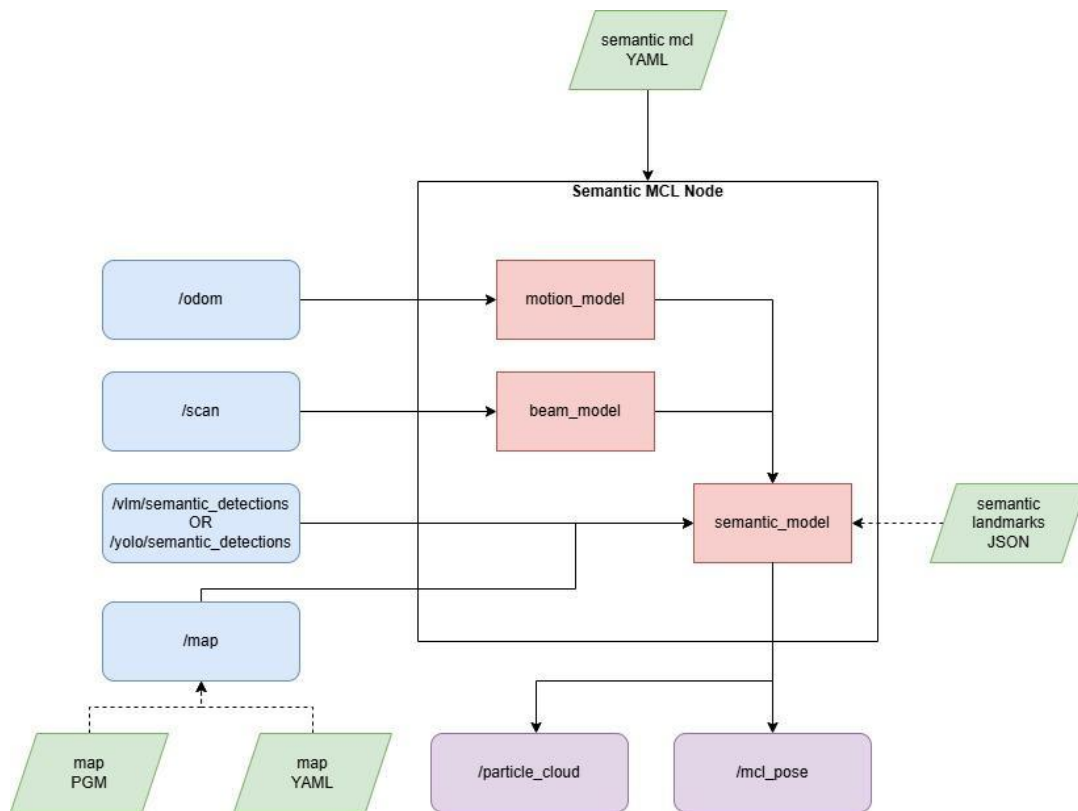
Based on a paper by Nicky Zimmerman (1), it does the following:

- Use of **CAD floor plans** for lidar scan of the environment's geometry
- Uses Monte-Carlo Localization (**MCL**) Framework to stochastically determine robot's pose on the map
- Uses a motion model that incorporates the **omnidirectional odometry**
- Combines **YOLO** semantic observation model with a lidar **beam** observation model to attenuate particles' weights across a map
- Uses a "**semantic map**," which indicates to the robot of where the semantic landmarks are on the map

Difference in Solution

- Zimmerman uses **room segmentation** to optimize particle distribution across the map during initialization. I have removed this step as the current operational environment is corridor, although future plans might include corridor segmentation
- Zimmerman uses the **YOLO** model for the semantic observation model, whereas I use the **VLM** Llava3 model for the semantic observation model.

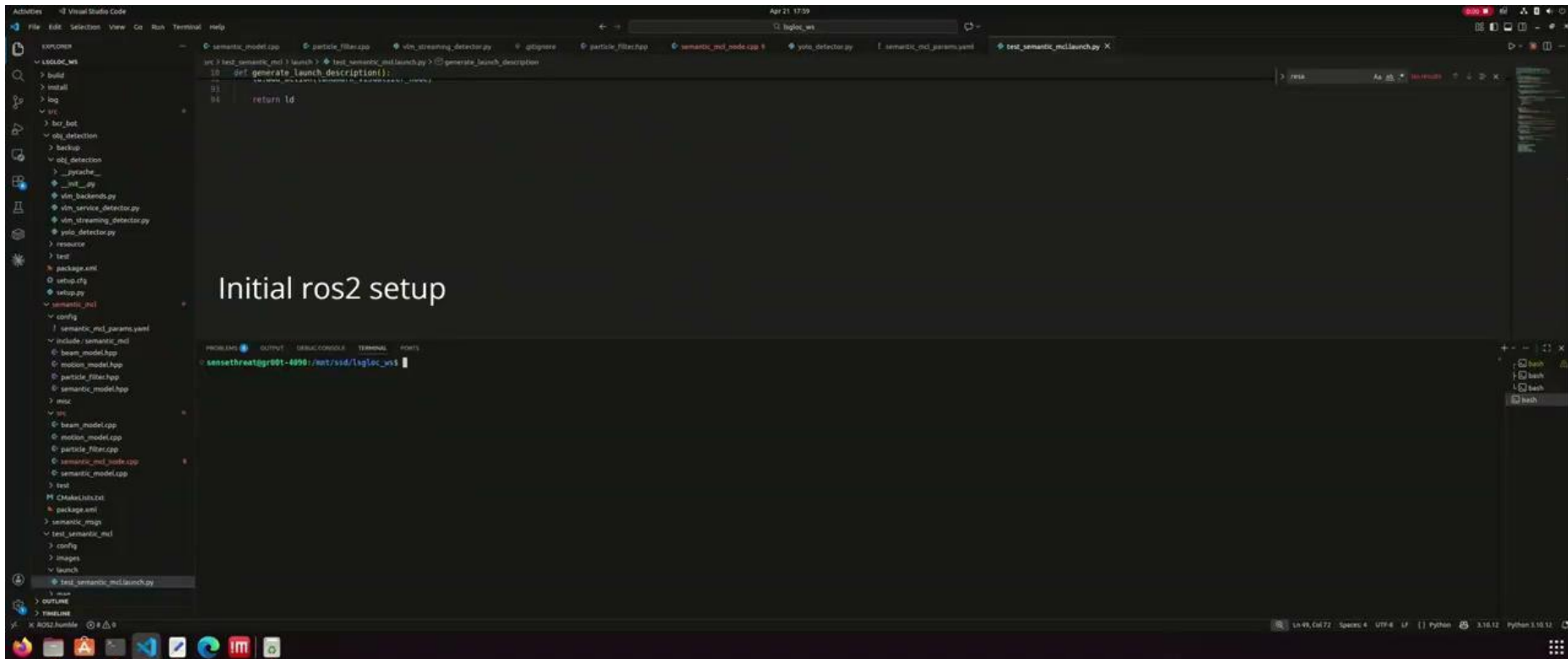
System Architecture



Global Localization Process

- 1) Particles are first **uniformly distributed** across the map with equal weights, i.e., equal likelihood that the robot is everywhere on the map
- 2) During initialization, the robot first inputs lidar and semantic observations to get the **best pose estimate** of where it is at.
- 3) As the robot moves, it takes in **more lidar and semantic observations** to get a better estimate of where it is at. It also takes into account its **odometry** and samples a new pose for each time step.
- 4) The more **unique** the semantic detections are, the better the candidate pose is at corresponding to the ground truth location.
- 5) The particles are **resampled** everytime a semantic detection occurs or when the weight values of the particles are similar to one another, i.e., uncertainty

Video Demo of LSGLOC



Limitations & Challenges

- **Map Format Requirement:** The robot must be given a map in PGM (only black and white pixels) and YAML format in order to do Monte Carlo Localization (MCL).
- **Map Scaling:** The lidar distances to nearest obstacles must correctly match in RViz2 or other software description applications.
- **Semantic Map Requirement:** The robot must be given a “semantic map” that lists down the locations of semantic landmarks on the map.

Limitations & Challenges

- **Global Localization Efficiency:** Given an environment with a limited number of semantic landmarks the robot can recognize, the robot will take a longer time to converge to the right location on the map.
- **VLM Hallucinations:** If we use VLM as our primary object detection model, the VLM “might” inaccurately describe objects in the scene.
- **VLM Latency:** If we use VLM as our primary object detection model, the VLM will take time to process the input payload and produce semantic detections.

Limitations & Challenges

- **Dynamic Object Interaction:** Currently, the semantic landmarks are assumed to be statically placed with respect to the semantic landmark file.
- **Object Instances & Uniqueness:** LSGLOC works well when there are many objects in the scene, and they are unique and different from one another.

Future Project Implementation

- 1) **Global Localization → Pose Tracking:** After the robot's pose has been correctly identified, the robot should be able to maintain a unimodal probability distribution over a long period of time.
- 1) **Localization Confidence Tracking:** The robot should indicate and quantify to the user how well it localizes with respect to the map and sensor observations without manual intervention.
- 1) **Path-Planning Integration:** The robot will be integrated with the Nav2 stack to perform waypoint navigation.

Possible Future Challenges during Deployment

- 1) **VLM Processing:** If the VLM is used as the primary object detection model, it might use a lot of power and computation, thus affecting the rest of the system.
- 1) **Sim2Real Gap:** Since the simulated robot model is different from the ones we have in real life, the odometry will be different and “might” be challenging to the localization. Dynamics has been filtered out of the simulation which affects the localization process even more.

Reference(s)

- 1) Zimmerman, N., Guadagnino, T., Chen, X., Behley, J., & Stachniss, C. (2022). Long-Term Localization using Semantic Cues in Floor Plan Maps. ArXiv.org. <https://arxiv.org/abs/2210.01456>